# Apache JMeter Performance Testing Project

## Project Description:

**Apache JMeter Performance Testing Project - E-Commerce Application**

- Designed and executed Load, Stress, Spike, Endurance, and Volume Testing using Apache JMeter.
- Created 50 performance test cases covering Login, Search, Cart, Checkout, Payment, and Order modules.
- Analyzed Response Time, Throughput, TPS, and Error Rates using Aggregate and Summary Reports.
- Identified performance bottlenecks and provided optimization recommendations.
- Generated performance reports and documented findings using industry-standard practices.

## Module 1: Home Page Performance

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM001 | Load Home Page with 50 users | 50 VU | Response time < 2 sec |
| JM002 | Load Home Page with 100 users | 100 VU | No errors |
| JM003 | Load Home Page with 500 users | 500 VU | System remains stable |
| JM004 | Verify throughput | 100 VU | Throughput increases proportionally |
| JM005 | Verify homepage under spike load | 50→500 VU | No crashes |

## Module 2: User Login Performance

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM006 | Login with valid users | 100 VU | Successful login |

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| JM007 | Login API response time | 200 VU | < 3 sec |
| JM008 | Login under peak load | 500 VU | Error rate < 2% |
| JM009 | Concurrent login requests | 1000 VU | System available |
| JM010 | Authentication service stress test | Increasing load | Bottleneck identified |

## Module 3: Product Search Performance

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM011 | Search product keyword | 100 VU | Results displayed quickly |
| JM012 | Search multiple products | 200 VU | Stable response |
| JM013 | Search with filters | 300 VU | Filter response within SLA |
| JM014 | Search special characters | 100 VU | No application failure |
| JM015 | Search API spike test | 100→1000 VU | Service remains available |

## Module 4: Product Details Page

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM016 | Open product page | 100 VU | Response < 3 sec |
| JM017 | Load product images | 200 VU | Images load correctly |
| JM018 | View specifications | 300 VU | No timeout |
| JM019 | Product recommendation section | 300 VU | Loads successfully |
| JM020 | Product page endurance test | 1 hr | Stable performance |

## Module 5: Shopping Cart

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM021 | Add item to cart | 100 VU | Success |
| JM022 | Add multiple items | 150 VU | Cart updates correctly |
| JM023 | Remove item from cart | 100 VU | Response acceptable |
| JM024 | Update quantity | 200 VU | Correct calculations |
| JM025 | Cart endurance testing | Continuous load | No memory leaks |

## Module 6: Wishlist

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM026 | Add product to wishlist | 100 VU | Success |
| JM027 | Remove wishlist item | 100 VU | Success |
| JM028 | Retrieve wishlist | 200 VU | Fast response |
| JM029 | Add same product repeatedly | 100 VU | No duplicates |
| JM030 | Wishlist stress test | 500 VU | Service stable |

## Module 7: Checkout

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM031 | Open checkout page | 100 VU | Response < 3 sec |
| JM032 | Select address | 200 VU | No delay |
| JM033 | Add new address | 150 VU | Data saved |

| | | | |
|---|---|---|---|
| JM034 | Verify order summary | 200 VU | Accurate data |
| JM035 | Proceed to payment | 300 VU | Success |

## Module 8: Payment Performance

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM036 | UPI payment | 100 VU | Payment successful |
| JM037 | Credit card payment | 100 VU | Response < 4 sec |
| JM038 | Debit card payment | 100 VU | Success |
| JM039 | Payment cancellation | 100 VU | System recovers correctly |
| JM040 | Payment stress test | 500 VU | Gateway remains available |

## Module 9: Order Processing

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM041 | Order placement | 100 VU | Order created |
| JM042 | Order confirmation page | 100 VU | Loads successfully |
| JM043 | Order email trigger | 100 VU | Notification generated |
| JM044 | Inventory update | 200 VU | Inventory adjusted |
| JM045 | Multiple order requests | 500 VU | No duplicate orders |

# Module 10: Overall System Performance

| TC ID | Scenario | Load | Expected Result |
|---|---|---|---|
| JM046 | Load test | 500 Users | Stable performance |
| JM047 | Stress test | 1000 Users | Bottleneck identified |
| JM048 | Spike test | 100→1000 Users | No crashes |
| JM049 | Endurance test | 4 Hours | No degradation |
| JM050 | Volume test | Large dataset | Performance acceptable |

**Load Test (100 Users):**

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|---|---|
| Average Response Time | 1.5 – 2.5 sec |
| Throughput | 80 – 150 req/sec |
| Error Rate | <1% |
| CPU Usage | 50 – 70% |
| Memory Usage | 3 – 6GB |
| Status | Pass |

**Stress Test (500 Users):**

| | |
|---|---|
| Average Response Time | 3 - 5 sec |
| Throughput | 200 – 400 req/sec |
| Error Rate | 2 - 5% |
| CPU Usage | 80 – 90% |
| Memory Usage | 6 – 8GB |
| Status | Pass/Warning |

**Spike Test (100 →1000 Users):**

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|---|---|
| Average Response Time | 5 - 8 sec |
| Error Rate | 5 - 10% |

| | |
|---|---|
| CPU Usage | 90%+ |
| Observation | Temporary degradation observed. |

**Endurance Test (4 Hours):**

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|---|---|
| Average Response Time | 2 - 2 sec |
| Error Rate | <1% |
| Memory Leak | Not Observed |
| Status | Pass |